CS4433 – Software Engineering

Individual Project

Project Management Plan

Moderate Learning Management System

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Course: Software Engineering

Development Model: Waterfall with V-Model Verification and Validation

Feasibility Study

The feasibility study asks if the proposed Moderate Learning Management System (LMS) can be built, used and completed successfully. We look at five things: technical, operational, economic, legal and schedule feasibility. We also consider the seven-week Waterfall development process.

Technical Feasibility

Technical feasibility asks: can we build the LMS with the technology we have?

The LMS will be a C++ application. It will use C++17 and SQLite to store data. It will run on one school computer or on the school's local network. It does not need a website or a mobile application. The project will also use CMake, Git, a testing tool such as Google Test or Catch2, and a code checking tool such as cppcheck or clang-tidy.

One important problem is that the school's computers are old and the internet is not reliable. The IT officer says that the system must work offline. Therefore, the system should not need the internet to work. SQLite is suitable for storing the data locally, and C++17 can be used to build the system.

The system must have 4 types of users: Administrator, Students, Lectures and Head of Departments. Each user must be allowed to do the things that their role allows them to do. The system must also store data in SQLite, use safe database queries, protect passwords, keep a record of changes to marks, check user input and handle errors safely.

The system must also work with information for up to 800 students. Therefore, we must test the system with large amount of data and test it on the school's computers before it is deployed.

Overall, the system is technically possible, but we must test it on the school's computers.

Operational Feasibility

Operational feasibility asks: can the school actually use the LMS?

The LMS has four main users and each user has a different job. Students need to review their courses and assignments, submit work, resubmit work and see their release marks and feedbacks. Lectures need to create assignments, set deadlines, give extensions, mark work and give feedback. Heads of the department need to check and approve marks before students can see them. The administrator manages users, departments, courses and student enrollments.

The system must be easy to use because the principal says that teachers are not IT experts. The system should therefore have simple steps and clear menus. Users should not need advanced skills to use it.

There is also a disagreement in the case study. The lecturer wants students to see their marks as soon as the lecturer finishes marking. However, the Principal says that students must not see their marks until the Head of Department has checked them.

Therefore, the system must include a moderation step before marks are shown to students. Overall, the LMS is operationally possible if it is simple to use, follows the agreed school process and gives users the help and instructions they need.

Economic Feasibility

Economic feasibility asks: Can the project be built and used without unreasonable costs?

The project will be a C++ application running on one school computer or the local network. It will not need a website or mobile application. It also does not include online payment, video conferencing, automatic marking or plagiarism detection. This means that the project does not need many extra services or systems. C++17 and SQLite are also suitable for this type of system.

However, the assignment does not give us a project budget. It also does not tell us how much new hardware, staff or training would cost. Therefore, we should not make up any money amounts.

The main costs that may need to be considered are:

- the school's computers;
- possible hardware upgrades;
- development and testing;
- staff training;
- deployment; and
- future maintenance

Before the system is used in a real school, the school should check that it has enough money, suitable computers and enough people to support the system.

Based on the information we have, the project is economically reasonable as an academic project. However, a real school deployment would need an approved budget and confirmed costs.

Legal Feasibility

Legal feasibility asks: Can we build and use the LMS while following the law?

The LMS will store personal and school information about students and staff. Therefore, this information must be protected.

Lesotho's Data Protection Act, 2012 deals with the use and protection of personal information. The system should therefore only collect information that is needed for the LMS. It should also protect this information and allow the correct handling of information that users may need to access or correct.

The system must also have good security. This includes controlling who can access information, protecting passwords and protecting student marks and other assessment information.

The LMS is for a secondary school, so the system will also deal with children. Lesotho's Children's Protection and Welfare Act, 2011 includes protections for children, including matters related to education, non-discrimination and children with disabilities. The system must therefore be designed with the safety and needs of learners in mind.

Accessibility is also important. The Persons with Disability Equity Act, 2021 supports inclusive education and recognises the needs of people with disabilities. Therefore, the LMS should be designed so that people with disabilities can use it as far as possible.

The project is legally possible if these legal requirements are considered when we design the system, protect the data and create the user interface. Legal requirements should also be checked again before the system is deployed.

Schedule Feasibility

Schedule feasibility asks: Can we finish the whole project in the seven weeks we have?

The project has a fixed time of seven weeks.

The planned schedule is:

- Week 1: Planning and management
- Weeks 1–2: Requirements
- Weeks 2–3: System design
- Weeks 4–5: Implementation
- Weeks 5–6: Testing
- Week 6: Deployment
- Week 7: Maintenance and completion

The different parts of the project depend on each other. For example, we need to understand the requirements before we can finish the design. We also cannot start implementation until the design has passed Gate 2. Testing needs a working system, and deployment depends on successful testing.

Therefore, the seven-week schedule is possible, but it is tight. We must carefully watch the schedule, complete important tasks on time, manage risks and make sure that each phase is completed before the next dependent phase begins.

If requirements or design take too long, the delay could affect the rest of the project because the project follows the Waterfall model.

Feasibility Recommendation

After looking at the technical, operational, economic, legal and schedule issues, the proposed LMS is possible to build and use as an academic software project.

Work Breakdown Schedule and Structure

Work Breakdown	WORK	TASKS
1	Project planning	Feasibility, work Breakdown, schedule, risk management
2	Requirement engineering	Stakeholder analysis, conflict resolution, SRS, use cases
3	System design	Architecture, UML, patterns, database
4	Implementation	C++ development, SQLite, security, CI
5	Verification and validation	Unit, integration, system and acceptance testing
6	Configuration	Git, CI, static analysis, metrics, code review
7	Deployment	Deployment diagram, installation guide, manuals, v1.0
8	Maintenance	CR1, impact analysis, v1.1, regression testing
9	completion	Professional/legal/ethical issues, final report, viva

RISK REGISTER

This is how I worked:

Likelihood and Impact

1 = very low

2 = low

3 = medium

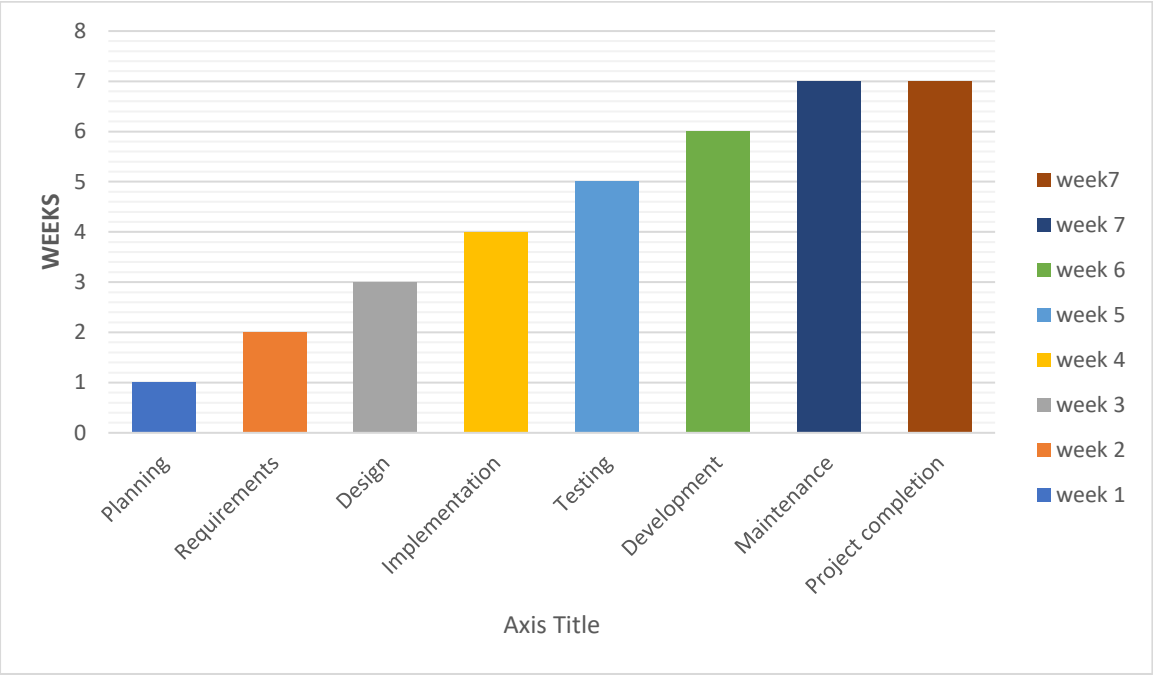
4 = high

5 = very high

Exposure = Likelihood * Impact

Risk ID	Risk	Likelihood	Mitigation	Impact	Exposure	Owner	category
1	Old school computers may not run the application adequately	**4**	Optimize the application and test early	**4**	**16**	Developer	Technical
2	Unreliable Network may affect the systems operation	**4**	The system will be designed to work offline	**4**	**16**	Developer	technical
3	Requirements conflicts may require rework	**4**	Resolve stakeholder conflicts before requirements are baselined	**4**	**16**	Project manager	project
4	Users may struggle to use the system	**3**	Keep workflows simple and provide role-specific user manuals and training	**4**	**12**	Project manager	Project
5	Stakeholders may disagree about late-penalty or extension rules	**3**	Resolve business-rule conflicts during requirements engineering and document agreed rules	**4**	**12**	Project manager	Project/people
6	Insufficient testing time may leave defects unresolved	**3**	Start automated testing early	**5**	**15**	Developer	Project

Gantt Chart


8
7
6
5
4
3
2
1
0
WEEKS
Planning
Requirements
Design
Implementation
Testing
Development
Maintenance
Project completion
Axis Title
week7
week 7
week 6
week 5
week 4
week 3
week 2
week 1